Project Report

CS-304 Database Systems

Sports League Management System

Submitted By:

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Submitted to:

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Declaration

We hereby declare that this project report entitled “Sports League Management System” is our own work. It has been carried out under the guidance of our instructor as part of the CS-304 Database Systems course. This report is submitted in partial fulfillment of the requirements for BS Cyber Security, 4th Semester, HITEC University, Taxila. Any material taken from other sources has been properly acknowledged.

Acknowledgement

We would like to express our sincere gratitude to [Instructor Name] for the guidance and support provided throughout the development of this project. Their feedback was valuable in shaping the structure and quality of our work. We are also thankful to our group members and classmates for their cooperation and assistance during this project.

Abstract

This project presents the Sports League Management System, a web-based application built using React, TypeScript, and a relational data model to manage the day-to-day operations of a multi-team sports league. The system organizes information across twenty interconnected modules, covering coaches, teams, players, venues, seasons, fixtures, match results, statistics, and fan-facing operations such as tickets and sponsorships. It provides complete CRUD (Create, Read, Update, Delete) functionality for every module, enforces relational integrity through foreign key references, and applies business-rule validation to prevent inconsistent data entry. The system was developed as an academic requirement for the CS-304 Database Systems course at HITEC University, Taxila, to demonstrate practical understanding of relational database design and full-stack application development.

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1. Introduction

Sports leagues involve a large amount of interconnected information — teams, players, coaches, venues, fixtures, results, and finances all need to be tracked accurately and kept consistent with one another. Managing this information manually, or through disconnected spreadsheets, quickly becomes error-prone as the number of teams, seasons, and matches grows.

The Sports League Management System is a web-based application developed to address this problem. It allows a league administrator to manage every aspect of league operations from a single dashboard, including team rosters, match scheduling, live standings, player statistics, ticketing, and fan registrations, while keeping all related records consistent through a relational data model.

This project was developed as an academic requirement for the CS-304 Database Systems course at HITEC University, Taxila, to apply relational database design concepts — including primary keys, foreign keys, referential integrity, and data validation — in a practical, full-stack application.

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2. Objectives

The main objectives of this project are as follows:

- To design and implement a relational data model with twenty interconnected tables
- To apply foreign key relationships and maintain referential integrity across modules
- To implement complete CRUD operations for every module through a unified web interface
- To enforce field-level and cross-field validation rules to maintain data quality
- To manage teams, players, coaches, venues, and seasons within one connected system

- To track match schedules, live results, standings, and player statistics
- To manage league commercial operations such as tickets, sponsors, and fan registrations
- To demonstrate practical database design and full-stack development concepts from the CS-304 course

3. Scope of the Project

The scope of this project covers the development of a complete sports league management system spanning twenty data modules. It includes team and player management, venue and season management, match scheduling and live results, standings and statistics tracking, and league commercial operations such as ticketing, sponsorships, posters, and fan registration. Every module supports full Create, Read, Update, and Delete operations along with search functionality.

The system is limited to a single-administrator setup with no multi-role access control. It does not include real-time match broadcasting, online payment processing, or public-facing fan portals, as these were considered beyond the scope of the academic requirement.

4. Tools and Technologies

The following tools and technologies were used to develop this project:

- React 18 with TypeScript — frontend framework for building the user interface
- Vite — build tool and development server
- Tailwind CSS — utility-first styling framework for responsive design
- Browser localStorage — client-side persistence layer for the in-memory database

- Visual Studio Code — code editor used for development
- Node.js / npm — package management and local development environment

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5. System Design

The system follows a component-based single-page application architecture. A central data store manages all twenty tables in memory and persists them to the browser's localStorage, while a schema definition layer describes the fields, validation rules, and display columns for each module. The user interface is generated dynamically from this schema, so every module shares the same consistent CRUD behaviour.

5.1 System Architecture

The application is structured around three core layers: a data layer (store.ts) that defines the database shape, seed data, and validation logic; a schema layer (schema.tsx) that defines field types, dropdown options, and table columns for each of the twenty modules; and a presentation layer (App.tsx and components.tsx) that renders the sidebar navigation, data tables, and modal forms based on the active module.

5.2 Project Structure

The project is organized into a small number of focused files rather than one file per module. The schema-driven design means that adding or modifying a module only requires updating its definition in the schema layer, and the table view, form modal, and validation logic update automatically.

5.3 Data Persistence

The system uses the browser's localStorage as its persistence layer. On first load, the database is seeded with sample data covering all twenty modules; subsequent changes are saved automatically after every create, update, or delete operation, so data survives page refreshes.

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6. Database Design

The data model is organized into twenty tables, each representing a distinct entity within the sports league domain. Each table has a unique numeric identifier and is connected to related tables through foreign key fields, allowing data such as team names and player names to be resolved and displayed wherever they are referenced.

6.1 Database Tables

The following twenty tables are included in the data model:

- Coaches — stores head coach details including name, contact information, specialty area, and years of experience
- Teams — stores team details linked to a head coach, including founding year and home city
- Players — stores player details linked to a team, including position, age, and jersey number
- Venues — stores stadium/venue details including address, seating capacity, and surface type
- Seasons — stores season details including start and end dates, year, and status
- Schedules — stores fixture details linking home team, away team, venue, and season
- Matches — stores match results linked to a schedule, including status and attendance
- Scores — stores quarter/half-wise scoring records linked to a match and team
- Game Logs — stores in-match events such as goals, assists, and cards linked to a player
- Standings — stores league table data including wins, losses, draws, and points per team per season
- Statistics — stores season-wise player performance statistics such as goals and assists
- Player Profiles — stores extended biographical data for each player such as height, weight, and nationality
- Injuries — stores player injury records including severity and expected return date

- Awards — stores award records linked to a player and a season
- Equipment — stores team equipment inventory including quantity and condition
- Sponsors — stores sponsor contract details including contract period and amount
- Posters — stores promotional poster details linked to a match
- Tickets — stores ticket types, pricing, and sales data linked to a match
- Fan Registration — stores registered fan details including their favourite team
- References — stores cross-table reference notes used for internal record linking

6.2 Key Relationships

The following foreign key relationships exist in the data model:

- teams.coach_id references coaches.id
- players.team_id references teams.id
- schedules.home_team_id and schedules.away_team_id reference teams.id
- schedules.venue_id references venues.id
- schedules.season_id references seasons.id
- matches.schedule_id references schedules.id
- matches.home_team_id and matches.away_team_id reference teams.id
- scores.match_id references matches.id; scores.team_id references teams.id
- gamelogs.match_id references matches.id; gamelogs.player_id references players.id
- standings.team_id references teams.id; standings.season_id references seasons.id
- statistics.player_id references players.id; statistics.season_id references seasons.id
- playerprofiles.player_id references players.id
- injuries.player_id references players.id
- awards.player_id references players.id; awards.season_id references seasons.id
- equipment.team_id references teams.id
- posters.match_id references matches.id

- tickets.match_id references matches.id
- fanregistration.favorite_team_id references teams.id

6.3 Sample Data

The system is seeded with sample data covering all twenty modules, including five coaches, six teams, twelve players, five venues, four seasons, eight scheduled fixtures, and corresponding matches, scores, standings, and statistics, giving a realistic dataset of over one hundred records to demonstrate the system's relational behaviour.

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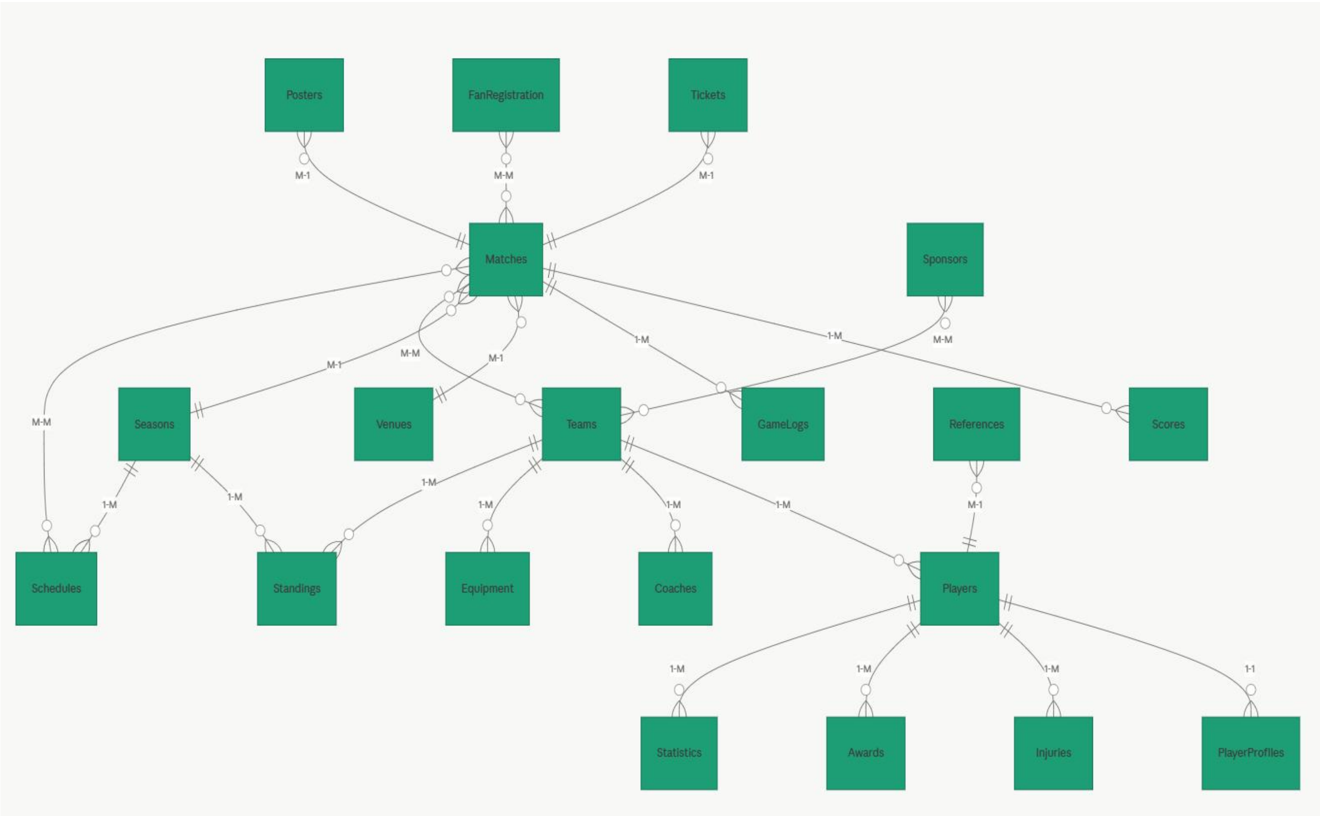
7. ERD



8. Implementation

The system was implemented in phases. First, the data model and validation rules were designed and all twenty tables were defined with their fields and relationships. Next, a generic CRUD table component and form modal were built to work with any module based on its schema definition. Finally, the dashboard, search functionality, and cross-field validation rules were added to complete the system.

8.1 Data Layer

The data layer is implemented in store.ts using a custom React hook (useDB) that manages the in-memory database state, synchronizes it with localStorage, and exposes create, update, remove, and reset operations to the rest of the application.

8.2 CRUD Operations

Every module supports Create, Read, Update, and Delete operations through a shared table-and-modal interface. The Add button opens a form modal pre-populated with the module's fields; submitting the form validates the data, applies the change, and displays a success notification before closing the modal. Edit and Delete follow the same pattern, with Delete requiring confirmation before the record is removed.

8.3 Validation

Field-level validation enforces required fields, minimum lengths, numeric ranges, email format, and letters-only constraints depending on the field type. In addition, cross-field validation rules are applied for specific modules — for example, ensuring the home and away teams in a fixture are not the same, that a season's end date is after its start date, and that tickets sold cannot exceed tickets available.

8.4 Search Functionality

Every module includes a live search bar above its data table. The search matches against both raw field values and resolved foreign key labels, so a search term can match either text content (such as a team or player name) or numeric values (such as an ID, jersey number, or year), allowing administrators to quickly locate records using either characters or numbers.

9. Results and Output

The system was tested locally using the Vite development server. All twenty modules were verified to work correctly, with data creation, editing, deletion, and search functioning as expected across the full dataset.

The dashboard successfully displays summary counts for all twenty modules along with recent matches, top standings, and top scorers, giving administrators an at-a-glance view of league activity.

The following modules were tested and verified as fully functional:

- Dashboard with live module counts, recent matches, top standings, and top scorers
- Coach, team, and player management with add, view, edit, and delete operations
- Venue and season management linked across fixtures and standings
- Schedule and match management with status tracking and attendance records
- Score and game log tracking linked to individual matches
- Standings and statistics modules reflecting season-wise team and player performance
- Player profile, injury, and award management linked to individual players
- Equipment, sponsor, poster, ticket, and fan registration modules for league operations
- Cross-module search functionality by both name (characters) and ID/numeric values
- Full CRUD lifecycle with success/update/delete confirmation notifications

10. Screenshots

10.1 Dashboard Overview

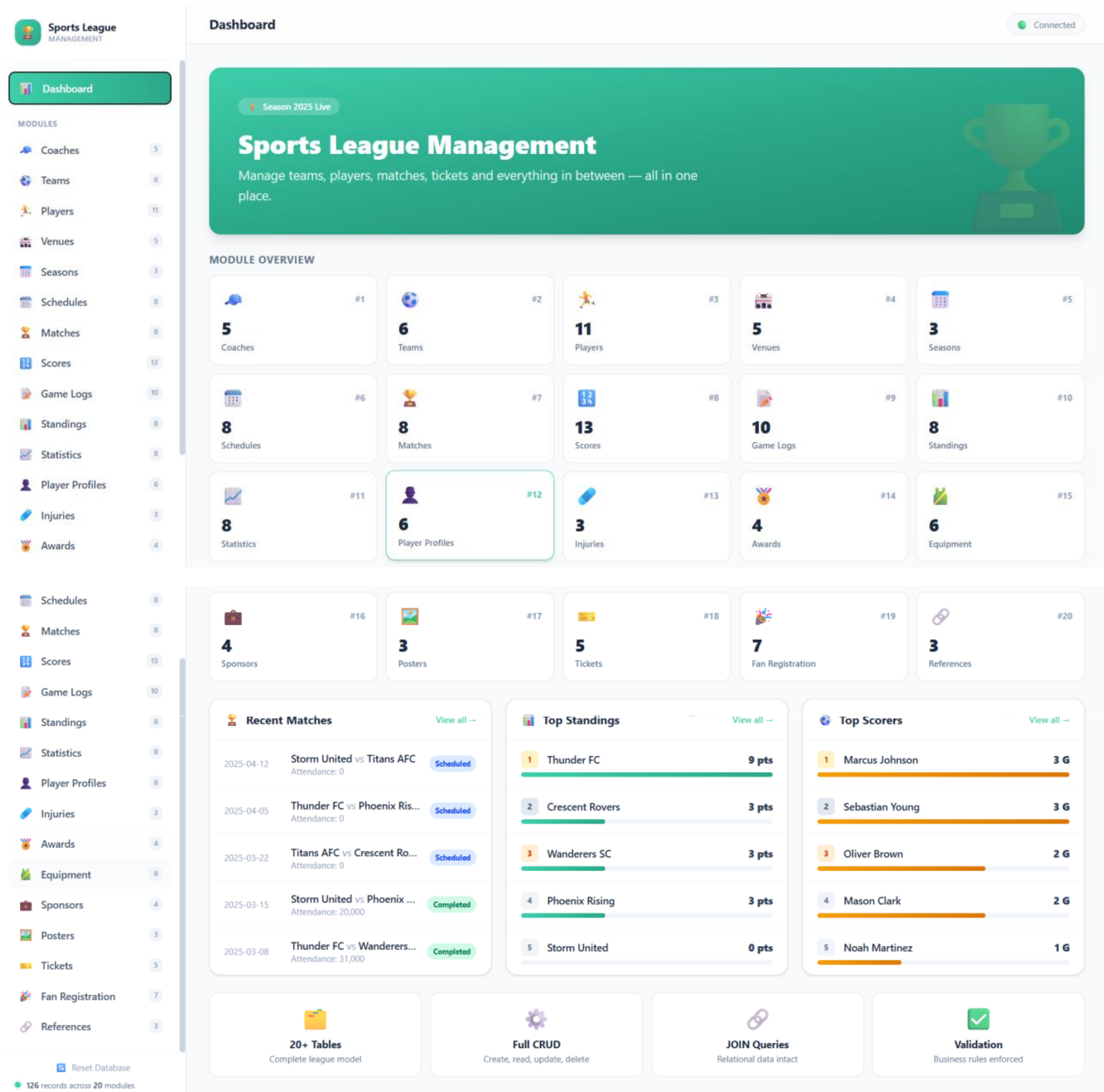


Figure 1: Dashboard showing all the Tables, Recents and all other details

10.2 Teams Module

Teams 6

Search... + Add Team

ID	NAME	COACH	FOUNDED	CITY	ACTIONS
1	Thunder FC	Carlos Martinez	2005	Downtown	
2	Storm United	David Thompson	2008	Westside	
3	Wanderers SC	Alessandro Rossi	2001	Midtown	
4	Titans AFC	James O'Connor	2010	Harbor	
5	Phoenix Rising	Lucas Silva	2015	Hills	
6	Crescent Rovers	Carlos Martinez	2003	Eastside	

Figure 4: All Teams page showing records with search and action options

10.3 Players Module

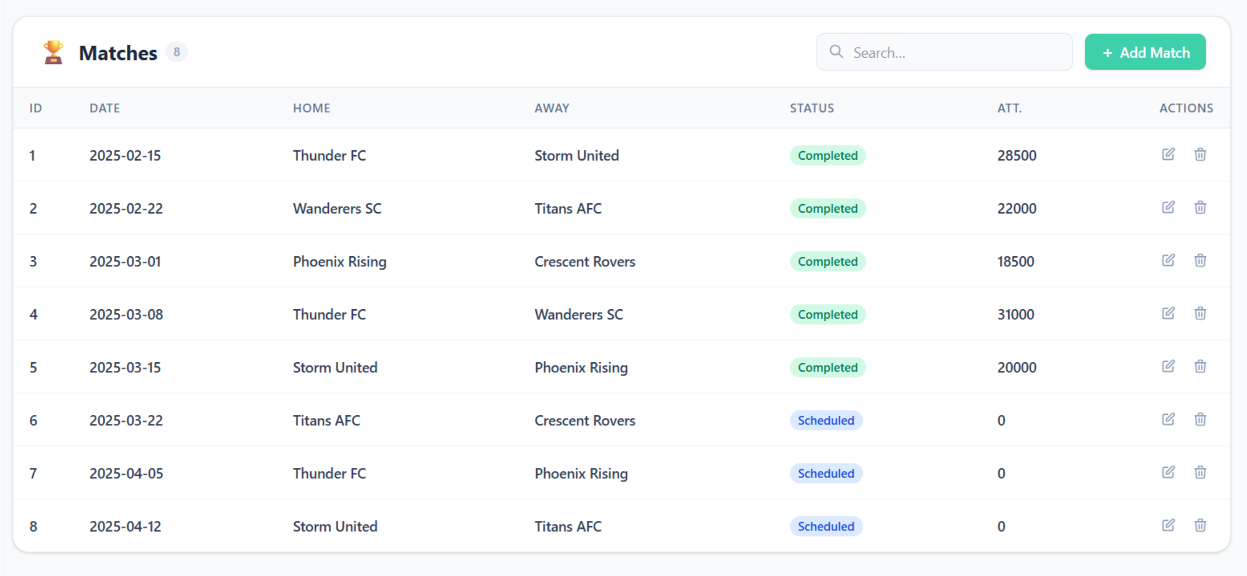
Players 11

Search... + Add Player

ID	NAME	TEAM	POSITION	AGE	#	ACTIONS
1	Marcus Johnson	Thunder FC	Forward	24	9	
2	Ethan Williams	Thunder FC	Midfielder	27	8	
3	Liam O'Brien	Storm United	Defender	29	4	
4	Noah Martinez	Storm United	Forward	22	11	
5	Oliver Brown	Wanderers SC	Midfielder	26	10	
6	Ava Garcia	Wanderers SC	Defender	25	3	
7	Isabella Lee	Titans AFC	Goalkeeper	28	1	
8	Mason Clark	Titans AFC	Forward	23	7	
9	Lucas Walker	Phoenix Rising	Midfielder	30	6	
10	Mia Hall	Phoenix Rising	Defender	21	5	
11	Sebastian Young	Crescent Rovers	Forward	24	9	

Figure 5: All Players page showing records with search and action options

10.4 Matches Module



Matches 8

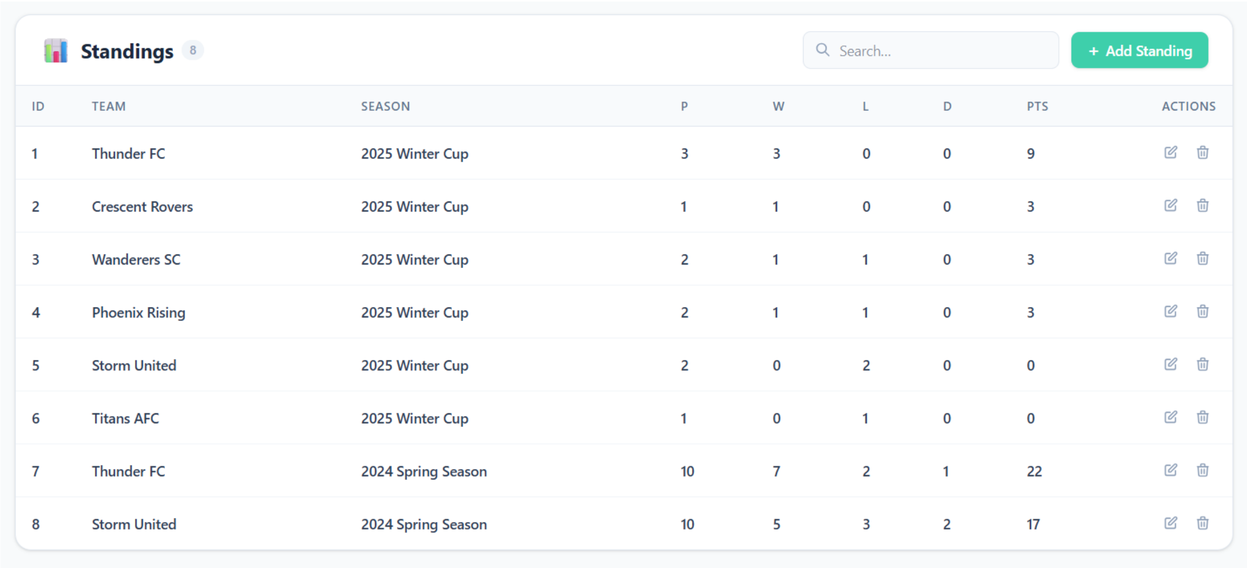
Search...

+ Add Match

ID	DATE	HOME	AWAY	STATUS	ATT.	ACTIONS
1	2025-02-15	Thunder FC	Storm United	Completed	28500	
2	2025-02-22	Wanderers SC	Titans AFC	Completed	22000	
3	2025-03-01	Phoenix Rising	Crescent Rovers	Completed	18500	
4	2025-03-08	Thunder FC	Wanderers SC	Completed	31000	
5	2025-03-15	Storm United	Phoenix Rising	Completed	20000	
6	2025-03-22	Titans AFC	Crescent Rovers	Scheduled	0	
7	2025-04-05	Thunder FC	Phoenix Rising	Scheduled	0	
8	2025-04-12	Storm United	Titans AFC	Scheduled	0	

Figure 9: All Matches page showing records with search and action options

10.5 Standings Module



Standings 8

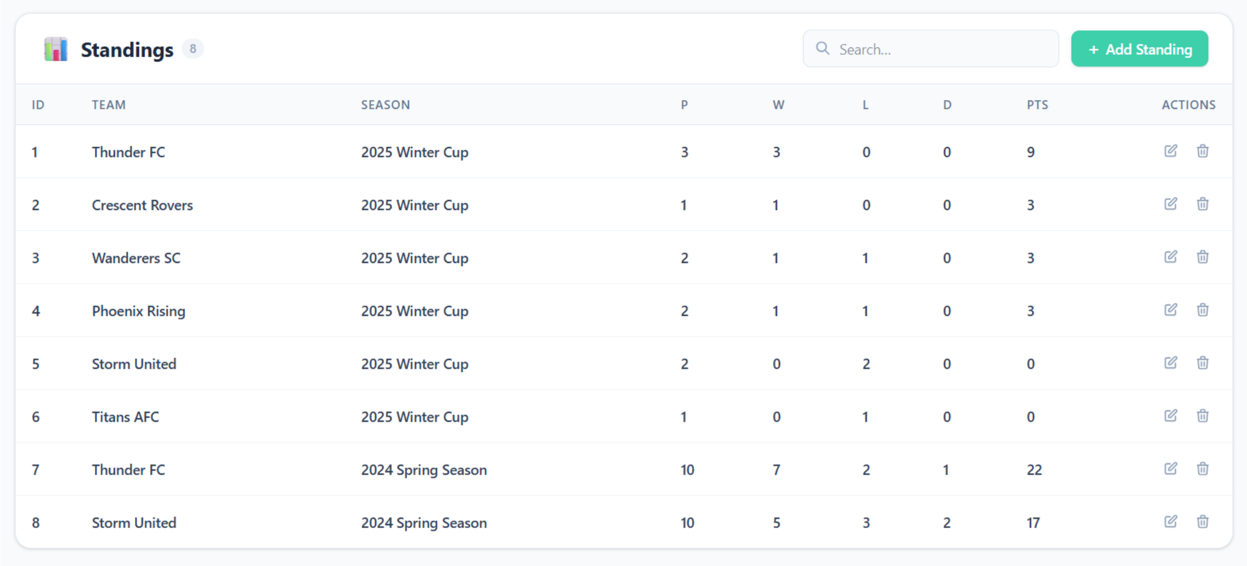
Search...

+ Add Standing

ID	TEAM	SEASON	P	W	L	D	PTS	ACTIONS
1	Thunder FC	2025 Winter Cup	3	3	0	0	9	
2	Crescent Rovers	2025 Winter Cup	1	1	0	0	3	
3	Wanderers SC	2025 Winter Cup	2	1	1	0	3	
4	Phoenix Rising	2025 Winter Cup	2	1	1	0	3	
5	Storm United	2025 Winter Cup	2	0	2	0	0	
6	Titans AFC	2025 Winter Cup	1	0	1	0	0	
7	Thunder FC	2024 Spring Season	10	7	2	1	22	
8	Storm United	2024 Spring Season	10	5	3	2	17	

Figure 12: All Standings page showing records with search and action options

10.6 Statistics Module



Standings 8

Search...

+ Add Standing

ID	TEAM	SEASON	P	W	L	D	PTS	ACTIONS
1	Thunder FC	2025 Winter Cup	3	3	0	0	9	
2	Crescent Rovers	2025 Winter Cup	1	1	0	0	3	
3	Wanderers SC	2025 Winter Cup	2	1	1	0	3	
4	Phoenix Rising	2025 Winter Cup	2	1	1	0	3	
5	Storm United	2025 Winter Cup	2	0	2	0	0	
6	Titans AFC	2025 Winter Cup	1	0	1	0	0	
7	Thunder FC	2024 Spring Season	10	7	2	1	22	
8	Storm United	2024 Spring Season	10	5	3	2	17	

Figure 13: All Statistics page showing records with search and action options

10.7 Player Profiles Module — CRUD Demonstration

The following sequence demonstrates the complete Create, Read, Update, and Delete lifecycle using the Player Profiles module as an example.

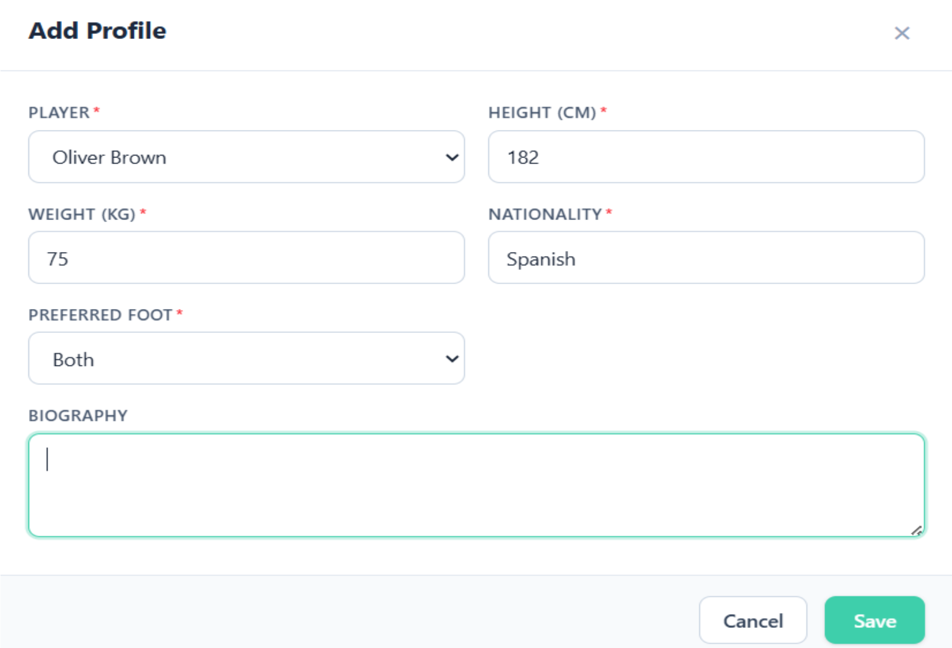


Figure 14: Add Profile form modal with player, height, weight, nationality, and preferred foot fields

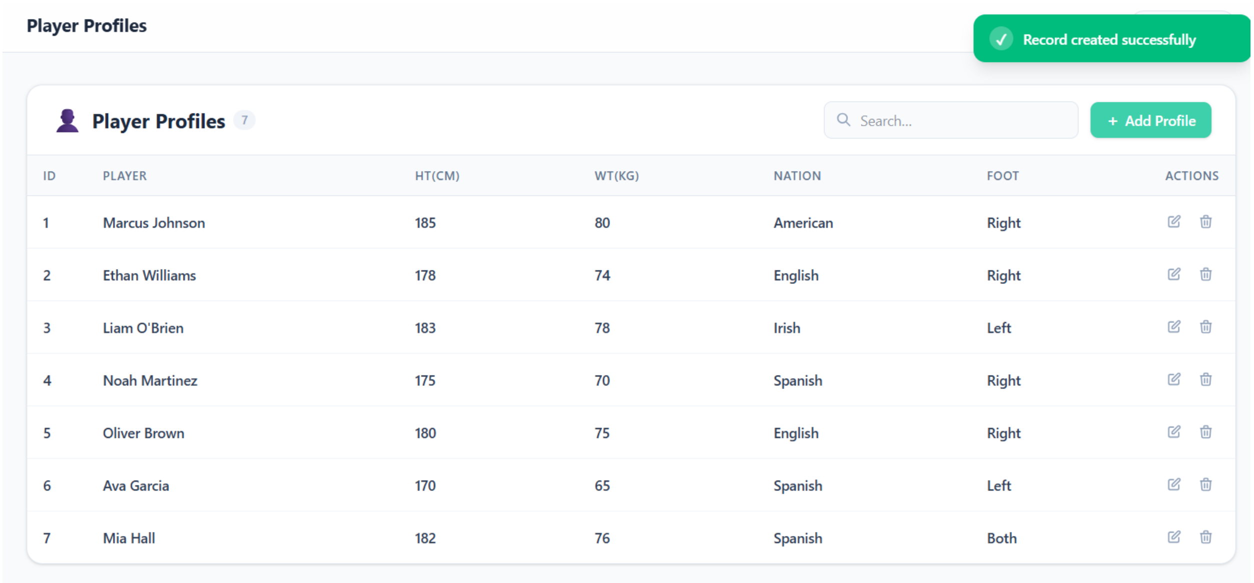


Figure 15: Player Profiles list showing "Record created successfully" confirmation after adding a record

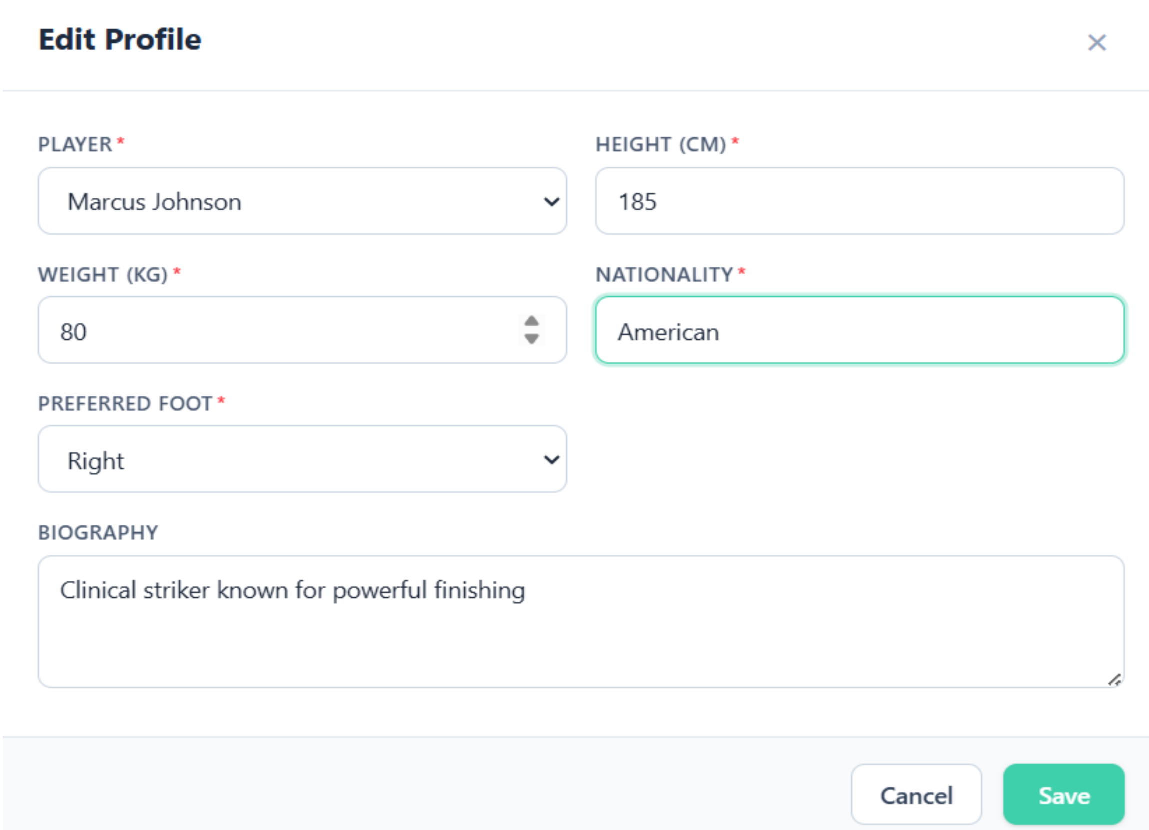
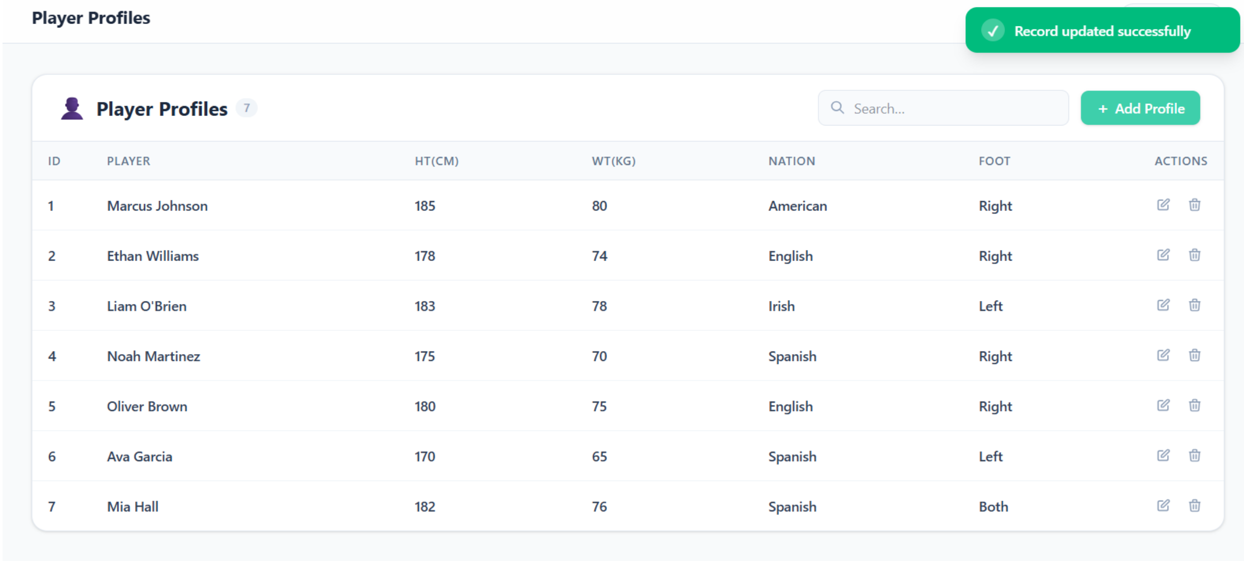


Figure 16: Edit Profile form modal pre-filled with the selected record's data



ID	PLAYER	HT(CM)	WT(KG)	NATION	FOOT	ACTIONS
1	Marcus Johnson	185	80	American	Right	
2	Ethan Williams	178	74	English	Right	
3	Liam O'Brien	183	78	Irish	Left	
4	Noah Martinez	175	70	Spanish	Right	
5	Oliver Brown	180	75	English	Right	
6	Ava Garcia	170	65	Spanish	Left	
7	Mia Hall	182	76	Spanish	Both	

Figure 17: Player Profiles list showing "Record updated successfully" confirmation after editing a record

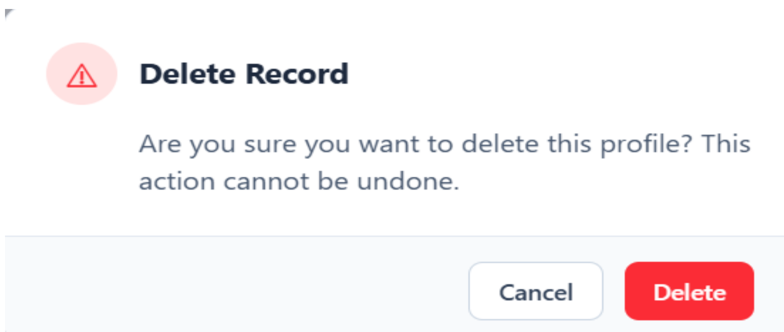
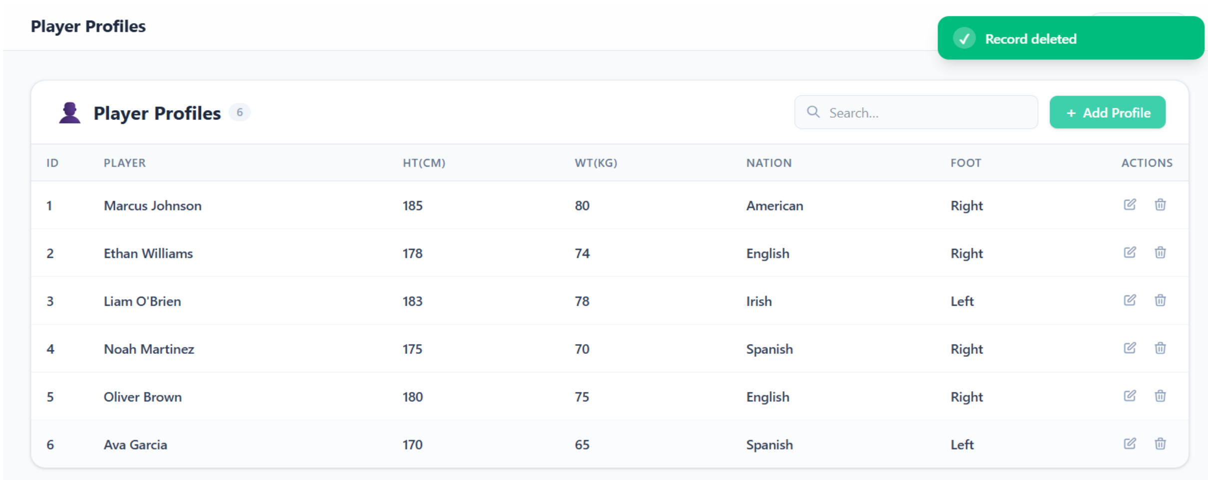


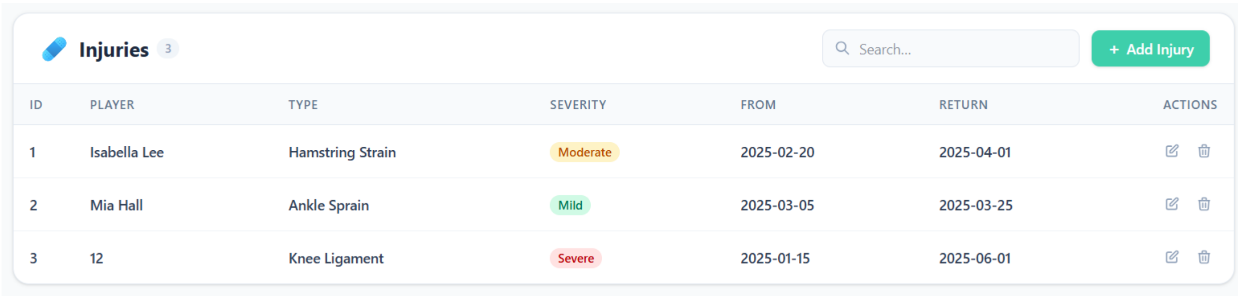
Figure 18: Delete confirmation dialog shown before a record is permanently removed



ID	PLAYER	HT(CM)	WT(KG)	NATION	FOOT	ACTIONS
1	Marcus Johnson	185	80	American	Right	
2	Ethan Williams	178	74	English	Right	
3	Liam O'Brien	183	78	Irish	Left	
4	Noah Martinez	175	70	Spanish	Right	
5	Oliver Brown	180	75	English	Right	
6	Ava Garcia	170	65	Spanish	Left	

Figure 19: Player Profiles list showing "Record deleted" confirmation after removing a record

10.8 Injuries Module


Injuries 3

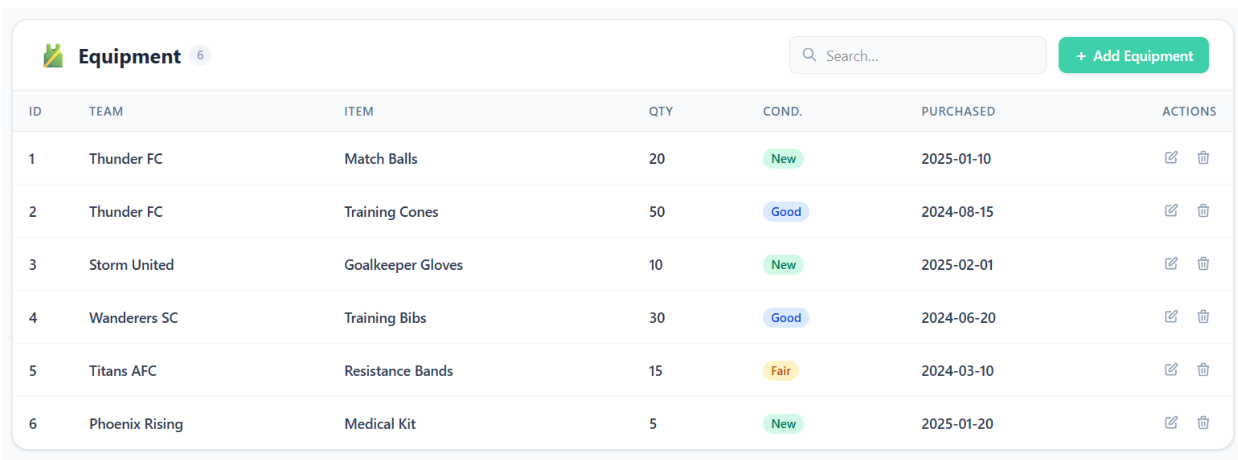
Search...

+ Add Injury

ID	PLAYER	TYPE	SEVERITY	FROM	RETURN	ACTIONS
1	Isabella Lee	Hamstring Strain	Moderate	2025-02-20	2025-04-01	
2	Mia Hall	Ankle Sprain	Mild	2025-03-05	2025-03-25	
3	12	Knee Ligament	Severe	2025-01-15	2025-06-01	

Figure 20: All Injuries page showing records with search and action options

10.9 Equipment Module


Equipment 6

Search...

+ Add Equipment

ID	TEAM	ITEM	QTY	COND.	PURCHASED	ACTIONS
1	Thunder FC	Match Balls	20	New	2025-01-10	
2	Thunder FC	Training Cones	50	Good	2024-08-15	
3	Storm United	Goalkeeper Gloves	10	New	2025-02-01	
4	Wanderers SC	Training Bibs	30	Good	2024-06-20	
5	Titans AFC	Resistance Bands	15	Fair	2024-03-10	
6	Phoenix Rising	Medical Kit	5	New	2025-01-20	

Figure 22: All Equipment page showing records with search and action options

10.10 Search Functionality

The search bar supports matching by both characters and numbers, allowing records to be located either by text fields (such as names) or numeric fields (such as IDs, jersey numbers, or years).

Players 7

T

+ Add Player

ID	NAME	TEAM	POSITION	AGE	#	ACTIONS
1	Marcus Johnson	Thunder FC	Forward	24	9	
2	Ethan Williams	Thunder FC	Midfielder	27	8	
3	Liam O'Brien	Storm United	Defender	29	4	
4	Noah Martinez	Storm United	Forward	22	11	
7	Isabella Lee	Titans AFC	Goalkeeper	28	1	
8	Mason Clark	Titans AFC	Forward	23	7	
11	Sebastian Young	Crescent Rovers	Forward	24	9	

Figure 28: Search results filtered using a text/character query

Venues 3

50

+ Add Venue

ID	NAME	ADDRESS	CAPACITY	SURFACE	ACTIONS
1	Olympic Stadium	123 Sports Ave, Downtown	50000	Grass	
2	Riverside Arena	456 River Rd, Westside	35000	Turf	
3	Central Park Field	789 Park Ln, Midtown	25000	Grass	

Figure 29: Search results filtered using a numeric query

11. Conclusion

The Sports League Management System was developed successfully as a CS-304 Database Systems project. It covers all major aspects of league operations through twenty interconnected modules, complete CRUD functionality, and enforced data validation.

This project helped us understand how a real-world relational system is designed and implemented, from defining entity relationships and referential integrity rules to building a consistent, schema-driven user interface that scales across many related data modules.

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12. Future Enhancements

The following enhancements can be added in future versions of the system:

- Backend database integration with a persistent server-side relational database
- Admin authentication and role-based access control
- PDF and Excel export for match reports, standings, and ticket sales
- Interactive charts and analytics on the dashboard
- Online ticket purchasing and payment gateway integration
- Automated standings calculation based on match results
- Email/SMS notifications for fans and registered users
- Mobile-responsive design improvements
- Cloud deployment for live, multi-user access
